

Programming Fundamentals (CS1002)

Course Instructor(s):

Dr. Faisal Cheema, Mr. Shehreyar Rashid,
Ms. Nirmal Tariq

Section(s): BS(CS)-25 (A,B,C,D,E,F,G,H,I)

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 70

Total Questions: 5

Date: Sep 23, 2025

251-0571

CS-B

@munir

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

CLO 1: Demonstrate the basic concepts of programming

Q1: The following C++ program is designed to check whether an input character is an uppercase alphabet, a lowercase alphabet, or not an alphabet at all. [Marks 10]

Task: Complete the missing conditions so that the program produces the correct output. Write your answers in the provided space only; answers outside the given space will not be accepted.

Restrictions:

1. Do not use the `<cctype>` or any other similar header file.
2. Do not use the relational operators `>=`, `<=`, or the logical operator `&&`.

```
#include <iostream>
using namespace std;
```

```
int main() {
```

```
    char ch;
```

```
    cout << "Enter a character: ";
```

```
    cin >> ch; // 2 marks
```

```
    if ( ch >= 'A' && ch <= 'Z' ) { // 4 marks
```

```
        cout << ch << " is an Uppercase Alphabet." << endl;
```

```
    }
```

```
    else
```

```
        if ( ch >= 'a' && ch <= 'z' ) { // 4 marks
```

```
            cout << ch << " is a Lowercase Alphabet." << endl;
```

```
        }
```

```
    else {
```

```
        cout << ch << " is Not an Alphabet." << endl;
```

```
    }
```

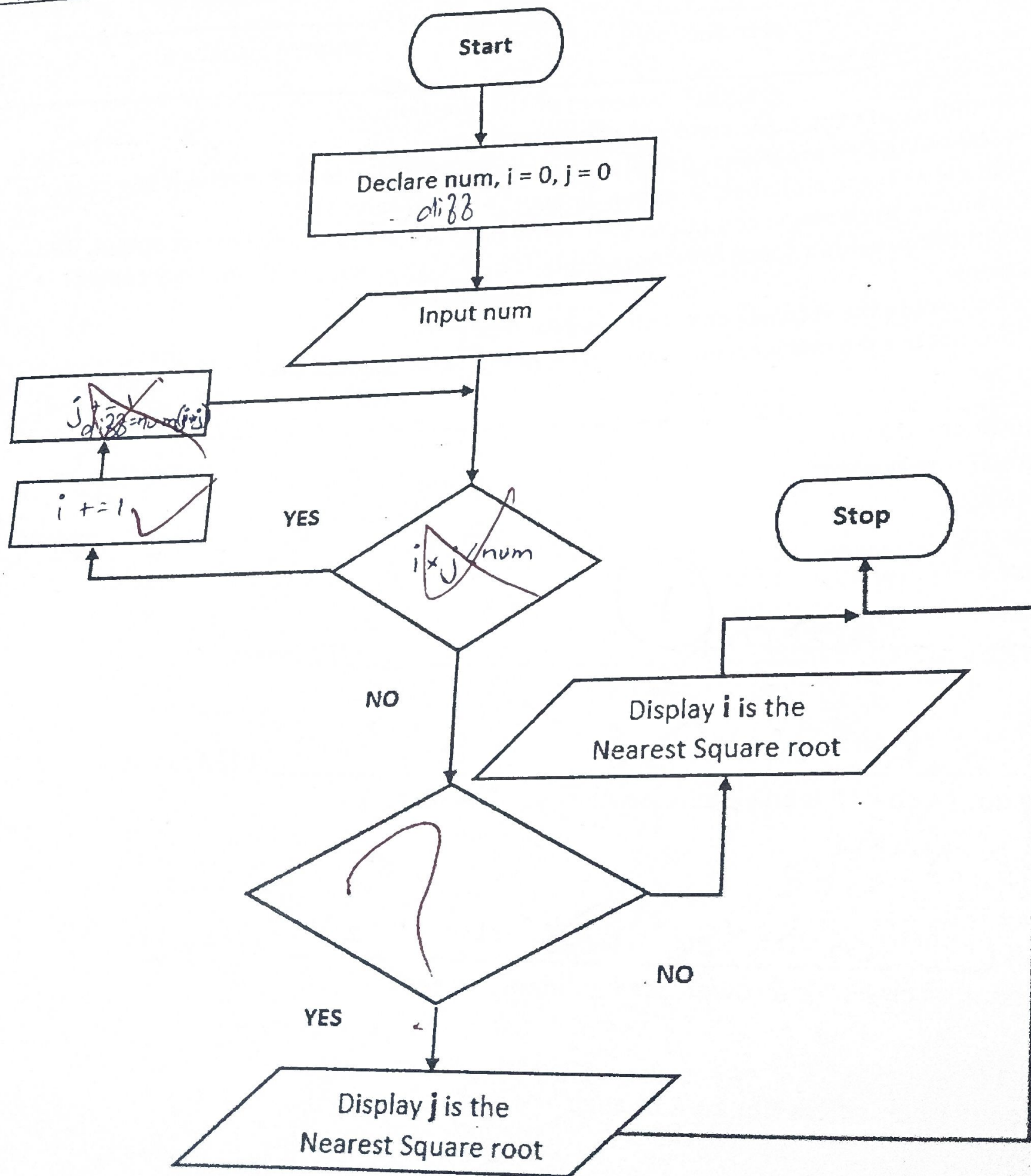
```
    return 0;
```

```
}
```

CLO 1: Demonstrate the basic concepts of programming

Q2: Complete the given flowchart by filling in the missing steps (2 process boxes & 2 decision boxes) so that it correctly finds the nearest square root ($\sqrt{}$) of a positive number without using built-in functions such as pow, sqrt, etc. Write your answers in the provided space only; answers outside the given space will not be accepted. [Marks 20]

Example 01	Example 02	Example 03
Input: num = 144	Input: num = 132	Input: num = 133
Output: 12 is the Nearest Square root	Output: 11 is the Nearest Square root	Output: 12 is the Nearest Square root



CLO 1: Demonstrate the basic concepts of programming

Q3: State the output of following code, assuming the code is error free. Write your answers in the provided space only; answers outside the given space will not be accepted. [Marks 10]

Code Segment:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double x;
    cout << "Enter a number: ";
    x=52;
    cout<<x;
    cout << "\nReport Card\n";
    cout << "=====\n";

    int base = static_cast<int>(x) % 7;
    double factor = x / 3 + base * 2 / 1.5;

    double bonus;
    if (!(x < 10)) bonus = factor * 2.5;
    else bonus = (factor + 5) / 2;

    double penalty = -1.5;
    if (bonus > 70)
        penalty = 4.0;

    cout << left << setw(12) << setfill('*')<<"Value"
         << right << setw(10)<<setfill('.') << fixed << setprecision(2) << x << endl;

    cout << left << setw(12) << setfill('*')<< "Factor"
         << right << setw(10) <<setfill('.')<< factor << endl;

    cout << left << setw(12) << setfill('*')<< "Bonus"
         << right << setw(10) <<setfill('.')<< bonus << endl;

    cout << left << setw(12) << setfill('*')<< "Penalty"
         << right << setw(10) <<setfill('.')<< penalty << endl;

    if (static_cast<int>(x) % 2 == 0)
        cout << "Note:\rEven number detected.";
    else
        cout << "Note:\\tOdd number detected.";

    cout << "\nPath: C:\\\\Users\\\\Public\t" << static_cast<int>(x + bonus) << endl;
}
```

 a b
a b

Output: Enter a number: 52.0
Report Card
=====
*****Value.....52.00
*****Factor33.33333333
*****Bonus 53.33333333
*****Penalty -1.5
Note: Even number detected.
Path: C:\\Users\\Public 195

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CLO 1: Demonstrate the basic concepts of programming

Q4: State the output of following code. If there is an error, write explicitly. If there is no output, write "No Output". Leaving output area as blank will result in 0 marks. No marks for partial correct output. Write your answers in the provided space only; answers outside the given space will not be accepted. [Marks 14 – 1 mark each]

	Code	Output
1	double x = static_cast<int>(7.7); int y = 3; cout << x % y;	1
2	cout << pow(2,3) % 2 << endl;	0
3	int easy = 3 * (-48 + 16 + 'a' / 2); cout << easy << endl;	48 ✓
4	int a = b = c = d = e = 10; a += b += c += d += e; cout << a;	50
5	short int s = 65536; if(s) cout << "True"; else cout << "False";	False ✓
6	unsigned short int u = 65792; char ch; <i>in local value random garbage value is stored</i> if(ch=u) cout << true; else cout << false;	Error: ch not initialized in line 2 in line ch==u is correct
7	unsigned char c = 250; c += 72; <i>322 - 256 = 66</i> cout << static_cast<char>(c) << endl;	B ✓
8	unsigned char ch = 888; cout << ch << endl;	X ✓
9	int a=5, b=15, c=4, d=1; a += b >> c > d; cout << a << endl;	5 ✓
10	int a = 5, b = 0; if (a = b) cout << pow(a,b); else cout << pow(b,a); <i>a becomes 0 so it is 0^0 = 1</i>	Logical error in line 2 should be a==b
11	int a=10, b=30; double c=37; b = c - b - a; <i>37-30-10 = -3</i> b -= 10; <i>→ b = -10 not b -= 10</i> cout << b << endl;	-13
12	double x = 5.5; cout << (static_cast<int>(x) % 3 + x * 2) << endl;	13 ✓
13	int a,b,c,d,e; a = b = c*3 = d+5 = e = 10; a += 25; b -= 30; c /= 10; d *= 5; e %= 15; cout << "a=" << a << " b=" << b << " c=" << c << " d=" << d << " e=" << e << endl; <i>d+5=e we cannot assign - smth to d's because it is not var</i>	a=70 b=15 c=4 d=1 e=0
14	int a = 25; cout << !(!(a));	1 ✓

- 32768 - 0 - 32767

Q5: State the output of following code. If there is an error, write explicitly. If there is no output, write "No Output".
Leaving output area as blank will result in 0 marks. No marks for partial correct output. Write your answers in the provided space only; answers outside the given space will not be accepted. [Marks 16 - 2 mark each]

Code	Output
<pre> 1 int x = 10, y = 5, z = 3, u = 4; if ((y--=2) < (x+=3) && (z+=1) <= y-1){ if (x > y (u+=1) <= y) cout << x+y*z-z*u << endl; cout << z/u << endl; cout << x << y << z << u << endl; } short int a = 65536; int b = (25/4); double c = pow(2.5, 2); a = a + b * c; double d = a / b; cout << d << endl; 2 int x = 1, y=2; if(y=0) cout << "A"; else if(y && (x=9)) cout << "B"; if(x>2); else cout << "D"; if(x<2 && y<x y == 0) cout << "E"; if(x<0); { cout << "A"; }</pre>	105 34 133 44 6 Logical Error Should be y==0 in Line 2 and x==9 in Line 4 New output: D old output: Bc DE
<pre> 4 int a = 10/3 + 18%5; int b = (a*5) / 7; int c = pow(2,3) + b; if (a > b && b != 0 && (a*b)/b == a) if (c % a == 0 (b - c) < 0) cout << a + b + c; else cout << c - a;</pre>	226
<pre> 5 char ch=128; if(ch>0) cout<<static_cast<int>(ch)<<endl; ch=-129; if(!(ch<0)) cout<<static_cast<int>(ch)<<endl; ch = ('a'-'A')*2+5; cout<<ch<<endl;</pre>	127 E
<pre> 6 char ch = 'd'; int val = ch - 'a'; int a = 14/3 + val; double z = a / 2.0; if (a % 2 == 1 && z > 3) cout << static_cast<int>(z * 2); else cout << a;</pre>	7
<pre> 7 double d = pow(3.0, 3) / 2; int x = static_cast<int>(d) - (5/2); if(x > 10 && d - x < 1.0)cout << x + static_cast<int>(d); else cout << x - static_cast<int>(d);</pre>	-2
<pre> 8 int marks = 85; if (marks >= 50) cout << "Grade D\n"; if (marks >= 60) cout << "Grade C\n"; if (marks >= 70) cout << "Grade B\n"; if (marks >= 80) cout << "Grade A\n"; if (marks >= 90) cout << "Grade A+\n"; else cout << "Failed" ;</pre>	Grade D Grade C Grade B Grade A Failed

End of Questions